PRATHIKA SHRI A.
CSE WIS CSDF.
W2 section.

SRM Reg. No.	R	A	2	4	1	2	7	0	3	0	1	0	0	2/3	9/8

SET F

SRM Institute of Science and Technology
College of Engineering and Technology
SCHOOL OF COMPUTING
ASSESMENT II

Course Code and Title: 21CSC203P / Advanced programming practice **Year / Sem: II / III**
Max. Marks: 25 **Duration: 1 Hr** **Date : 03.11.2025**

COURSE ARTICULATION MATRIX (CAM)

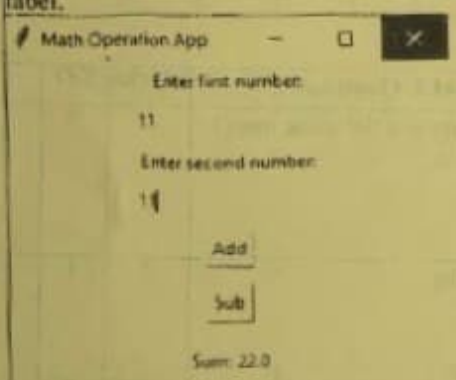
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	3	-	-	-	-	-	-	-	-	-	-
CO2	3	2	-	1	3	-	-	-	-	-	-	-	-	-	-
CO3	3	-	2	1	3	2	-	-	1	-	-	-	-	-	-
CO4	3	2	-	1	3	-	-	-	-	-	-	-	-	-	-
CO5	3	-	2	1	3	2	-	-	1	-	-	-	-	-	-

Sl. No	Part A (5* 2 = 10 Marks) (Answer ALL Questions)	Marks	CO	PO	BL	PI CODE
1.	The following code is meant to square all numbers in a list using map(), but it has an error. Identify and correct the mistake. numbers = [1, 2, 3, 4] result = map(lambda x: x*x, num) print(result)	2	4	1	1	4.1.1
2.	Complete the rule to find siblings using PyDatalog. from pyDatalog import pyDatalog pyDatalog._____ ('parent, sibling, X, Y, Z') sibling(X, Y) <= _____ (Z, X) & parent(Z, Y) & (_____ != _____)	2	4	3	4	4.3.2
3.	Write a short Python program using the multiprocessing module to create two processes that each print their process ID. Explain why this approach uses a distributed memory model.	2	4	3	3	4.3.1
4.	Write a Python program using the SymPy library to perform the following symbolic operations. Define a symbolic variable x and Compute and display the derivative of the expression $x^3 + 3x^2 + 3x + 1$	2	5	3	2	5.2.2
5.	Create a Tkinter program which is intended to display the message "Hello World!" in a label when the button is clicked.	2	5	3	3	5.2.2

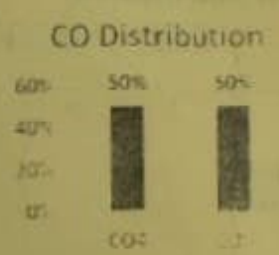
Part B (3* 5 = 15 Marks) (Answer ALL Questions)

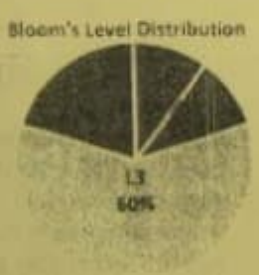
Sl.No		Question	Marks	CO	PO	BL	PI Code
Q6 a.	i)	You have the following list of employee salaries: salaries = [4500, 5200, 6100, 4800, 7000] Using a functional programming approach in Python, write a program to increase only the salaries above $5000 by 10% and display the updated list.	5	4	3	3	4.3.1
	ii)	What is a pure function in Python? Give an example.					
		(OR)					
Q6 b.	Suppose you have a family tree: • Mary is the parent of John and Jane.		5	4	3	3	4.3.4

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	• John is the parent of Tom. • Jane is the parent of Fred. Find out who are Tom's cousins.					
Q7 a.	Build a TCP client-server using Python sockets to send a string from client to server. Give client and server code snippets for the string exchange.	5	5	3	3	4.3.4
	(OR)					
Q7 b.	Write a Python program using the SymPy library to perform the following symbolic operations: • Define a symbolic variable x. • Compute and display the derivative, integral, and factorization of the expression $x^4 + 4x^3 + 6x^2 + 4x + 1$	5	4	3	3	4.3.3
Q8 a.	A student is creating a simple "Math Operation App" using Tkinter to perform addition and subtraction of two numbers entered by the user. Design a Tkinter window with two Entry boxes to accept two numbers and two buttons labeled "Add" and "Sub". When the button is clicked, display the sum and difference of the two numbers in a label. 	5	5	3	3	5.3.2
	(OR)					
Q8 b.	Write a Python program using the Automata library to simulate a DFA that accepts binary strings ending with '01'. Define states, transitions, input symbols, initial and final states. Test the DFA for the inputs: '01', '1001', '1010'.	5	5	1	3	5.3.1

Course Outcome (CO) and Bloom's level (BL) Coverage in Questions





R. Arcadia

Course Coordinator

Audit Professor

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